Haitian Zhu

Naperville, IL 60563 | (630) 362-6308 | haitiandzhu@gmail.com | hzhu50@uic.edu
linkedin.com/in/haitian-daniel-zhu | github.com/kiraguriz

Technical Skills

Languages: Java, Dart, TypeScript, JavaScript, C, C++, Swift, Go, Python, HTML/CSS, SQL
Frameworks & Tools: Flutter, React Native, Expo, Fastify, Node.js, JavaFX, Google Gemini AI SDK, ML Kit, Jest, JUnit, Xcode, Maven, Git, GDB
Concepts: Cross-Platform Mobile Development, REST API Integration, Real-Time Systems, AI/ML Integration, Object-Oriented Programming, Multithreading, Network Programming

Education

University of Illinois Chicago — Chicago, IL
B.S. in Computer Science — *Expected May 2027*
- **Relevant Coursework:** Data Structures, Computer Design, Software Design, Software Engineering, Programming Languages Design & Implementation, System Programming

Projects

OnlyPlanes — Live Flight Tracking App | *React Native, Expo, TypeScript, Fastify, SQLite* Apr 2026 – Present
- Co-designed a cross-platform aviation app (iOS & Android) with a team and solely implemented the entire codebase, a TypeScript monorepo spanning a React Native/Expo client, a Fastify backend, and a shared type package
- Built a real-time flight-tracking backend with Fastify and SQLite (WAL mode) that polls the OpenSky Network into an in-memory cache and serves REST endpoints for bounding-box queries, flight detail, and callsign/ICAO24 search
- Engineered an interactive live map (MapKit on iOS, Google Maps on Android) rendering directional aircraft markers with smooth position interpolation between ∼5s update intervals
- Architected a dark-first design system spanning 40+ screens (logbook, spotting locations, community forum) and set up GitHub Actions CI/CD enforcing lint, typecheck, build, and 50+ Jest test suites on every push

Workout Movement Tracker | *Flutter, Dart, SQLite, Google Gemini AI, ML Kit* Jan 2026 – May 2026
- Led development of a cross-platform fitness app (iOS & Android) in a 4-person team, authoring 42 of 91 total commits across database, AI, diet, goal, and UI layers
- Integrated Google Gemini AI to deliver five personalized coaching features (daily tips, workout suggestions, diet advice, post-session feedback, and weekly goal adjustments) with a file-based JSON caching strategy that reduced redundant API calls
- Architected a 15-repository SQLite data layer with schema migrations, supporting meals, workouts, body metrics, goals, and AI result persistence, plus a USDA FoodData Central integration with debounced search and offline-first caching
- Implemented an evidence-based goal engine using Mifflin-St Jeor and Schofield BMR formulas to auto-calculate daily calorie and macro targets, plus HealthKit/Google Fit sync for activity data

Multiplayer 3-Card Poker Game | *Java, JavaFX, Sockets* Nov 2025
- Developed a distributed client-server card game supporting multiple concurrent players with real-time state synchronization
- Engineered a robust multi-threaded server using TCP Sockets and Object Serialization to handle game states and client connections efficiently
- Implemented core game logic (Straight Flush, Hand Evaluation, dealer qualification) ensuring 100% accuracy through JUnit testing
- Designed a responsive GUI with custom CSS styling and MVC architecture to strictly separate game logic from the user interface

Chicago Traffic Violation Analysis System | *Python, SQL (SQLite)* Sep 2025
- Developed a console-based application to analyze and visualize Chicago traffic camera violation data using Python and SQL
- Constructed complex SQL queries utilizing inner joins, aggregations, and subqueries to extract insights on red light and speed camera violations
- Integrated SQLite with Python to dynamically retrieve data and implemented plotting functionality to visualize violation trends across city zones

Certifications

App Development with Swift Associate | *Apple* Issued Dec 2025